

Wireless Networking

Task 1 — Wi-Fi Standards

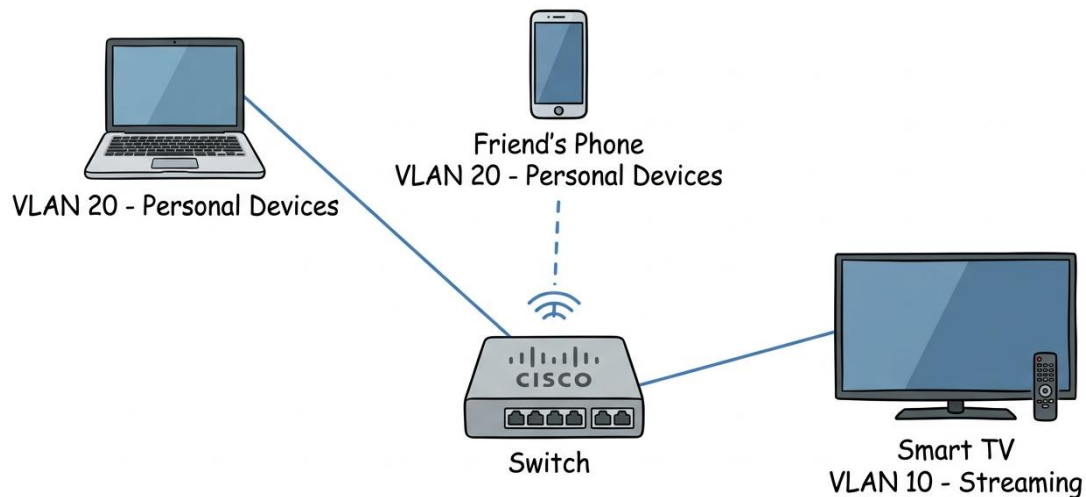
I found three popular Wi-Fi standards and their main differences.

Wi-Fi Standard	Frequency Band	Key Difference
802.11n (Wi-Fi 4)	2.4 GHz / 5 GHz	Supports speeds up to about 600 Mbps
802.11ac (Wi-Fi 5)	5 GHz	Provides higher speed than 802.11n
802.11ax (Wi-Fi 6)	2.4 GHz / 5 GHz	Better performance with many connected devices

Short Explanation:

Wi-Fi standards have improved over time. Newer standards provide better speed, efficiency and performance, especially when many devices are connected.

Task 2 — Smartphone to Internet Through Wi-Fi



Flow:

**Smartphone → Wi-Fi Router → ISP → Internet →
Website/Server**

The connection between my smartphone and Wi-Fi router is wireless. The router then uses a wired Ethernet connection to connect to the ISP and Internet.

Task 3 — Wi-Fi Network Scan

I scanned the available Wi-Fi networks using my device and noted their SSID, signal strength and security type.

No.	SSID	Signal Strength	Security Type
1	My Home WIFI	-45 dBm	WPA2
2	Home Network	-62 dBm	WPA3
3	Guest WIFI	-75 dBm	WPA2

Note: Replace these example values with your **actual Wi-Fi scan results** from your laptop/phone.

My Choice

I would choose the network with a **strong signal and good security**, preferably WPA3 if it is available. A stronger signal provides a more stable connection, while good security helps protect the network from unauthorized access.

Result:

The Wi-Fi scan helped me compare nearby networks based on signal strength and security.

Task 4 — SSID Hiding

SSID means **Service Set Identifier**. It is basically the name of a Wi-Fi network that we see when searching for available Wi-Fi networks.

SSID hiding means the Wi-Fi router does not normally broadcast its network name in the list of available Wi-Fi networks. A user needs to enter the SSID manually to connect.

Advantage

One advantage is that the Wi-Fi network name is not openly shown in the normal Wi-Fi network list, which can reduce casual visibility.

Disadvantage

One disadvantage is that hiding the SSID does **not provide strong security**. The

network can still be discovered with appropriate network tools, and users may find it less convenient to connect.

Conclusion

SSID hiding can reduce the visibility of a Wi-Fi network, but it should not be considered a replacement for strong Wi-Fi security such as WPA2 or WPA3.